

A pre-feasibility study for the production and distillation of medicinal and aromatic plants

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1. Executive summary

Aromatic plant oils are used as flavoring materials in food industries such as juices, soap, shampoo and detergents. It is also used in traditional medicine recipes as external use, as well as in the manufacture of some pharmaceuticals or flavoring materials in the food industry. The pre-desired varieties will be identified in cooperation with the pharmaceutical industry, detergents and cosmetics, pesticide industry, food industry and exporters.

The idea of this project is to establish a small plant specialized in extracting oils and extracts from aromatic plants. It is possible to start with the available plants in the area of some of the desired varieties or to cultivate them by farmers after the identification of the demand. In any case, if this plant distillated and filled its products in small bottles holding a capacity of 30-100 mm and sold them, to perfumery and herbal shops, it will achieve a good profit because the market is almost devoid of high quality products from the domestic industry.

Table 1 economical indicators

Indicator	
Total Investment Cost (Jordan Dinars)	126,000
Internal Rate of Return (IRR)	30.26%
Pay-Back Period (Years)	4
Net Present Value (Jordan Dinars)	273,000
Job Opportunities provided by the Project	7

The estimated capital of the project amounted to 126,342 JD, most of which are for construction works, site preparation (35,761), machinery and equipment (31,625), vehicles (16,500) and working capital (20,456). Where working capital is required for production inputs and workers' salaries. (50%) of the project will be financed through loans and the rest through self financing. Despite the high investment cost of the project, it will provide suitable employment opportunities commensurate with the size of the investment. The project will rely on exploitation of untapped raw materials and reliance on labor in manufacturing and marketing Essential oils of all kinds.

The results of the financial analysis indicate that the net value of the cash flows of the project is positive, which is anindicative of the financial feasibility of the project and the rate of internal revenue is much higher than the estimated cost rate of capital, which is indicative of the feasibility of the project. Other financial indicators indicate that the project is profitable to the investor.

Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

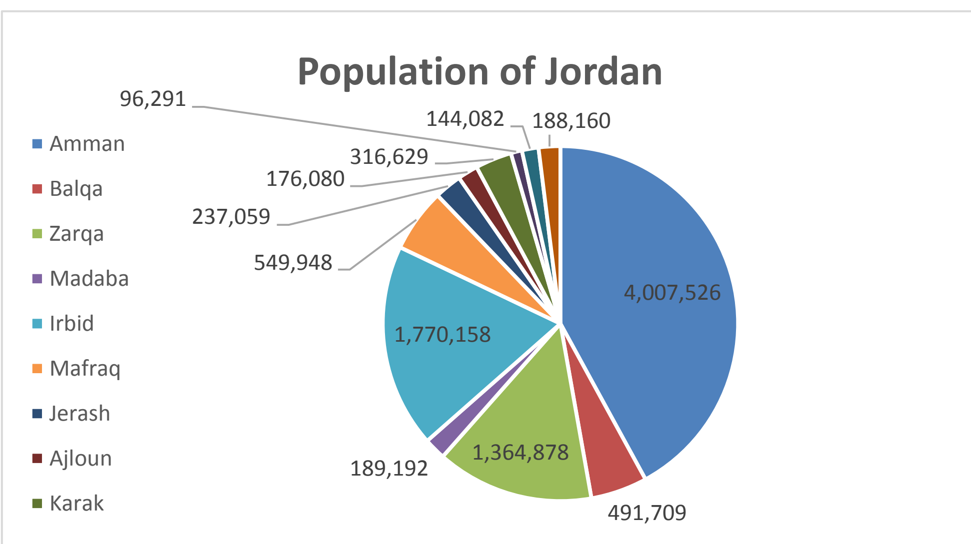
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table 2 Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	**%18.2**
15-64 Years	%61.02	%30.7	**%28.6**
More than 65 Years	%3.94	%1.6	**%1.6**

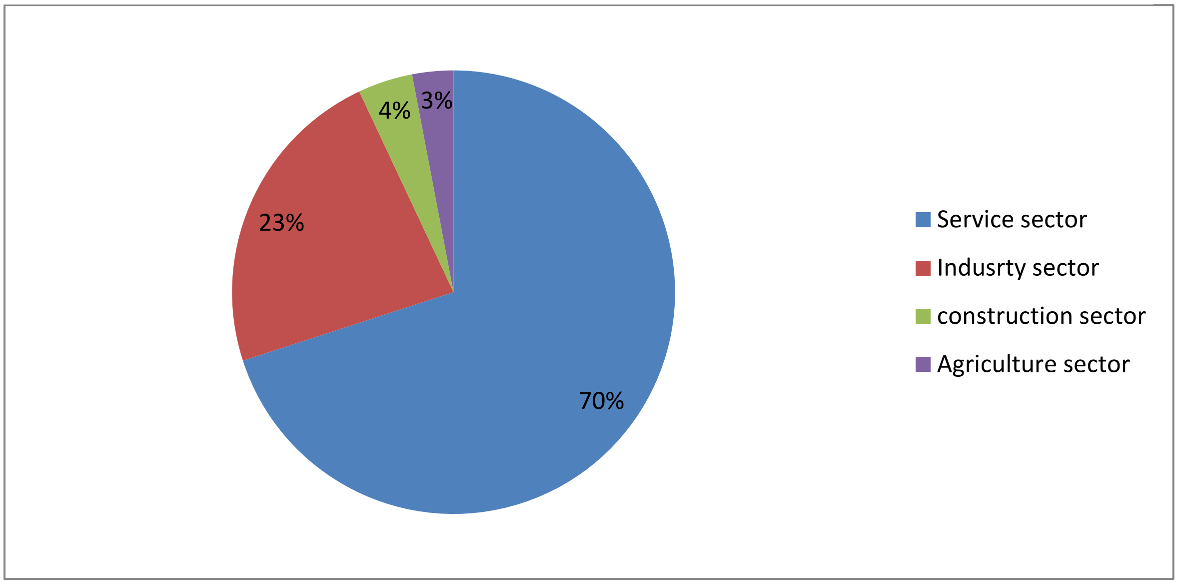
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2 Sectors Contibution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US$3.8 trillion market and compromising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce.

Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in it endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Invesment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

The investment environment in Ajloun:

As per in the Investment Law No. 30 of 2014 and the Income Tax Reduction Law in the Less Developed Areas (issued under Article (5) of the Investment Law No. (30) Of 2014), the following taxes and exemptions are imposed on investment projects:

Table 3 payable income tax and tax exemptions

Development areas	Less developed areas
Income tax is 5%	Value of income tax 20% According to the divisions in the law, Mafraq governorate is classified as category B where investors get 80% tax exemption on income tax for 20 years from the date of investment.

2. Market analysis

Project description

The pioneered projects depend on specialization and the use of technology in production, in addition to understanding the internal and external market requirements, and trying to provide the right product and rely on entering the market in time for profit. The achievement of profit depends on the optimal utilization of Jordan's climatic conditions, marketing support and the availability of agricultural expertise.

The project is based on the idea of covering the growing demand for medicinal and aromatic plant oils in the local markets and exporting to the neighboring markets by extraction of desired oils of plants of high quality and density according to specifications.

Oil and distillation of medicinal and aromatic plants are used in the manufacture of pharmaceuticals and medical drugs for the treatment of many diseases. They are used in various medical preparations as medically used as stimulants, antiseptics, tonics, therapeutic analgesics or drugs to reduce blood pressure and other medical uses.

Medicinal and aromatic plants are also used in the manufacture of cosmetics, detergents, soap and shampoo. It also extracts aromatic oils that are used in the manufacture of the finest international perfumes. An important use of medicinal and aromatic plants is its use as spices in food. As well as the preservation of canned food, sweets and soft drinks and others. These plants are used as flavor, aroma and food preservatives and extract the finest types of vegetable oils of various uses such as mint oil, lemon oil and orange oil. These are used in the manufacture of shampoos, soap and liquid fluids.

One of the important uses of medicinal and aromatic plant oils as natural insecticides is to kill or repel insects such as neem, and also to fight fungus and plant-damaging bacteria as well as diseases harmful to humans and animals and naturally harmless to the environment.

In Jordan, there are many aromatic plants such as Alkina and Qaisoum, and many aromatic plants such as roses, Rosemary, bitter orange (Abu Sfeir), and lemon. There is growing interest in the cultivation of aromatic and medicinal plants in Jordan, which are consumed in large quantities such as sage and thyme. The cultivation of aromatic and medicinal plants will be an additional source of income for farmers in rainfed areas such as Ajloun.

The project aims to diversify production and eliminate marketing problems. The project will produce aromatic oils of all kinds and supply the local market with various types of non-traditional aromatic oils.

Project objectives

There are increasing in quantities of medicinal and aromatic plant oils that available in Jordan imported from Pakistan and Turkey. The project aims to produce these oils for the local market in cooperation with detergents, soap, cosmetics, pharmaceuticals, pesticides, juices, food industries and spice and herbal dealers. The project aims to:

1. Achieving a profitable return to the owner of the project,
2. Production of good quality of medicinal and aromatic plant oils and their supply to the local market.
3. Supply part of the local demand of medical and aromatic plant oils
4. Production of oils and supply them to local industries that consume large amounts of them
5. Use available and unexploited local resources
6. Create job Opportunities in the less developed regions.

This project aims to extract medicinal and aromatic oils from various plants available in the Kingdom, especially in Ajloun Governorate, for its climate that suits these plants over the four seasons in order to benefit from them in many industrial, medical and nutritional uses. For food, drink or perfume as well as natural colored materials for some types of food

Proposed product

Oils and extracts of aromatic medicinal plants are important in many uses in industry such as:

- Food industries (natural flavorings and natural preservatives)
- Pharmaceutical industry (various types of medicines, eye dropsointments).
- Cosmetics industry (shampoo, cream, oils).
- Perfumery of all kinds.
- Chemical industries (soap, air freshener, insecticides).

There is a wide range of medicinal, aromatic and ordinary plants that aromatic oils can be extracted from them including mint, thyme, jasmine, lemon, orange, lavender, coconut, pineapple, sage, cloves, juniper, garlic, salts, almonds, roses, mustard, cider, flax, laurel leaf, Aloe, fenugreek, cinnamon and many other kinds of flowers and plants. The proportion of oils distilled from each type of plant varies.

Aromatic oils can be obtained from various plant parts such as leaves, flowers, fruits, stems, buds, roots, tubers, and lemals. Aromatic plants are present in the Kingdom, especially in Ajloun, such as thyme, mint, chamomile, aniseed, fenugreek, cloves, wormwood, turmeric and many others. The available plants in Ajloun:

1. Lemon Grass Extract:

Lemon grass is grown in the central valleys and can grow in different environments. Lemon grass contains stearal oil as an effective substance. The oil of lemon grass inhibits the growth of many fungi (eg, P. chrysogenum, A. fumigates, A.flavus macrophomina phasoli) and bacteria (Staphylococcus aureus, Bacillus subtillis, Escherichia coli, Pseudomonas aeruginosa, P.fluorescens).

The following ingredients are found in lemon grass powder (tunnis, alkaloids and glycosides), which play an important role in preventing the harmful effects of microbes. Therefore, the treatment of cowpea and corn with lemon grass powder prior to storage reduces the harmful effect of the above-mentioned microbes without affecting the vitality of these seeds, thus prolonging the storage period with the safe use of such crops. The extract of the lemon grass completely inhibits the growth of fungi (Ustilago maydis, Ustilaginoidea virens, Curvularia luntat, Rhizopusesp) and the use of lemon grass extract in the resistance of plant diseases caused by fungal infections (Botrytis cinerea, Rhizoctonia solani).

2. Basil extract:

One of the available herbaceous plants in Jordan. And the part used from this plant is the leaves, which contain oil with the substance of camphor and linalol, which is used as a gases repellent as well as in the perfume industry. Water and alcohol of basil extracts are used against plant pathogens such as fungus (A.alternata, Curvularia tuberculata). The basil extract inhibits mycelium growth as well as germination of F.oxysporum, which causes wilt disease for fenugreek. The basil extract is also used as an antimicrobial agent for the growth of many fungi loaded with seeds such as A.flavus, A.niger and F.moniliforme.

3. Peppermint oil - Mint:

Its one of the herbaceous plants grown in Jordan. Mint oil contains volatile oil of menthol, pethin and tannin and the part used is the leaves and floral peaks. The mint contains carotene oil with carvoons, lemons, pithin. The mint extract affects the growth of fungus such as Aspergillus nidulans and E. coli.

- Medical part: leaves and flowers.
- Active substance: leaf oil and volatile flowers.

4. Garlic extract:

The garlic extract resists many plant pathogens, especially the vegetative group. Garlic extract is used to resist many bacterial and fungal pathogens (Pseudomanas phaseclica, Xanthomonas sp., Puricularia oryzae, sp. Colletrotrichum, Pseudopernosporn cubnusis, and Monilia fructucola).

5. Anise oil

It is a herbaceous plant exists around the Mediterranean. It is now spread in most European countries. It is about half a meter high and the plant has smooth cylindrical shank, ribbed, and long branches with rounded leaves and the end of the branches, rounded compressed white headed flowers are being beared then transformed after maturation to a small floral-colored fruits and the plant lives one year.. A well-known plant of the Khayyim type. The part used from the plant is the fruit (seed) as well as the volatile oil only.

• Medical part: seeds.

• Active Ingredient: Aromatic oil containing Anethol

6. Chamomile oil:

Medical part: Flowers head.

• Composition: azulin, alcohol, epigptin, phytosterin acid, mineral salts, vitamins.

• Active ingredient: Azulen oil.

7. Parsley oil:

• Medical part: roots, leaves and seeds.

• Active ingredient: volatile oil called Apiol.

8. Violet oil:

• Medical part: leaves, flowers and pre-flowering roots.

• Composition: Aromatic oil containing iridine glycosides, nitroprobonyon saturic acid, methylsalicylic acid, kolid and ouduratine.

• Active ingredient: leaf oil, flowers and roots.

9. Watercress oil:

Medical part: Green and dried leaves.

• Composition: vitamins A and B, calcium, phosphorus, mineral salts (iodine, sulfur, iron).

• Active ingredient: leaf oil.

10. AlJadeh Oil:

• Medical part: papers.

• Active ingredient: leaf oil.

11. Thyme oil:

• Medical part: flowering branches.

• Active substances: volatile oil containing (5%) of Tamol and carboxolol and resin materials such as Rastan and tannin.

1. AlShumer oil:

• Medical part: seeds, roots and leaves.

• Active Ingredient: volatile oil containing (60%) of ethethol and about (17%) of finon and a small proportion of pine and anien.

13. Fenugreek oil:

• Medical part: mature seeds.

• Composition: trifonline, Collin, proteopene, oil, mineral salts, resin.

• Active ingredient: seeds oil.

14. Khubaisa oil:

• Medical part: leaves, shank and flower.

• Composition: Glues.

• Active ingredient: oil extracted.

15. Tulip Oil (White Lilies):

Medical part: Flowers.

• Active Ingredient: Oil contains sapolin.

16. Wormwood oil:

• Medical part: dried heads.

• Composition: Contains gonoxide of Santonin, Artisin and Giloxide tojon, Essentlin.

• Active ingredient: Santon oil.

17. Sage oil:

• Medical part: whole plant.

• Composition and active substances: Turbine Hydroxide, Mersin, Simine, Tannin.

18. Nettle oil:

• Medical part: whole herb including roots.

• Composition: Contains histamine, formic acid, hydrox, travertine, saccharine, iron, chlorophyll, mineral salts and C.A.

• Active ingredient: chlorophyll oil.

19. Aromatic water:

• Water emulsions and pure water solutions saturated with volatile oils.

• Used as a flavor enhancer such as rose water, peppermint water and others,

• It also have distinctive therapeutic qualities

It is proposed in this project to produce special extracts and oils from natural plants and diluted with alcohol or propyl glycol or oil with good concentrations which make the product preferable by the consumer and beneficial to the owner of the project, since 100% of pure extracts will be expensive for the consumer and may not be able to deal with them, so, it should be lightened when used.

Market size

It can be said that Jordan imports are about 200 tons of essential oils mixtures prepared for consumer and industrial use. These oils are extracted from many types of plants available in Jordan. These imports come under the terms of the customs tariff (item No. 3301) Of them) aromatic oils (though the turpents are removed) with or without wax, aromatic substances known as "Zerinoid" extracted oil resins, aromatic oils concentrated in grease, vegetable oils or candles, obtained by the method of sipping or soaking, secondary byproducts residues from the process of detenication of essential oils, distilled water and aromatic oils water solutions. The statistics of the Department of Statistics indicate that the import of aromatic oils of various kinds amount to about 210 tons annually as an annual rate for the period (2011-2015), and exports about 2016 tons, mostly of water solutions such as rose water. Table (4) shows the annual rate of Exports and imports of essential oils for the past five years.

The following tables show Jordan's imports of aromatic oils: orange, citrus, mint oil, other mint oil, and oils from non-citrus plants and other medicinal and aromatic oils

Table 3 Annual rate of quantity and value of exports and imports of aromaticoils (2011-2015)

Code	Description of classification	Imports: Jordanian Dinars	Imports: Quantity	National exports: JD	National exports: Quantity	Re-exported: Jordanian Dinars	Re-exported: Quantity	Total exports: Jordanian Dinar	Trade Balance (Total Exports - Imports): JD
330112000	Aromatic oils, from oranges	91,375	19,023	0	0	0	0	0	-91,375
330113000	Aromatic oils, from lemon	7,731	1,186	0	0	4,727	667	4,727	-3,004
330119000	Aromatic oils of other citrus	108,284	34,025	15,216	1,075	11,701	1,431	26,917	-81,367
330124000	Aromatic Oils of Mint (Menethapiereta)	21,124	1,295	0	0	5,604	1,792	5,604	-15,520
330125000	Aromatic oils of different kinds of mint	82,735	3,797	0	0	818	103	818	-81,916
330129000	Other non-citrus essential oils	392,011	56,436	12,607	5,532	50,909	11,741	63,516	-328,495
330190100	Aqueous solutions of aromatic oils prepared for medical purposes	34,522	12,018	96,844	194,843	22,550	2,973	119,394	84,872
330190900	Other aromatic oils	144,380	82,727	7,468	14,560	4,905	377	12,373	-132,007
	Total	882,161	210,506	132,136	216,009	101,213	19,083	233,349	-648,812

Source: Department of Statistics

Table 4 Evolution of exports and imports of aromaticoils, of oranges (330112000)

Year	Imports: Jordanian Dinars	Imports: Quantity	National exports: JD	National exports: Quantity	Re-exported: Jordanian Dinars	Re-exported: Quantity	Total exports: Jordanian Dinar	Trade Balance (Total Exports - Imports): JD
2011	79,426	9,898	0	0	0	0	0	-79,426
2012	165,007	41,079	0	0	0	0	0	-165,007
2013	55,528	13,918	0	0	0	0	0	-55,528
2014	59,034	14,148	0	0	0	0	0	-59,034
2015	97,880	16,072	0	0	0	0	0	-97,880
Rate	91,375	19,023	0	0	0	0	0	-91,375

Source: Department of Statistics

Table 5 Evolution of exports and imports of aromaticoils, of lemon (330113000)

Year	Imports: Jordanian Dinars	Imports: Quantity	National exports: JD	National exports: Quantity	Re-exported: Jordanian Dinars	Re-exported: Quantity	Total exports: Jordanian Dinar	Trade Balance (Total Exports - Imports): JD
2013	12,539	1,867	0	0	0	0	0	-12,539
2014	1,017	250	0	0	14,180	2,000	14,180	13,163
2015	9,637	1,440	0	0	0	0	0	-9,637
Rate	7,731	1,186	0	0	4,727	667	4,727	-3,004

Source: Department of Statistics

Table 6 Evolution of exports and imports of aromaticoils, of other citrus (330119000)

Year	Imports: Jordanian Dinars	Imports: Quantity	National exports: JD	National exports: Quantity	Re-exported: Jordanian Dinars	Re-exported: Quantity	Total exports: Jordanian Dinar	Trade Balance (Total Exports - Imports): JD
2011	52,185	12,144	0	0	0	0	0	-52,185
2012	142,456	35,080	18,586	743	0	0	18,586	-123,870
2013	111,777	33,491	0	0	796	390	796	-110,981
2014	97,256	38,806	14,972	1,505	14,400	2,000	29,372	-67,884
2015	137,744	50,603	42,523	3,125	43,307	4,763	85,830	-51,914
Rate	108,284	34,025	15,216	1,075	11,701	1,431	26,917	-81,367

Source: Department of Statistics

Table 7 Evolution of exports and imports of aromaticoils of Mint (330124000)

Year	Imports: Jordanian Dinars	Imports: Quantity	National exports: JD	National exports: Quantity	Re-exported: Jordanian Dinars	Re-exported: Quantity	Total exports: Jordanian Dinar	Trade Balance (Total Exports - Imports): JD
2013	12,539	1,867	0	0	0	0	0	-12,539
2014	1,017	250	0	0	14,180	2,000	14,180	13,163
2015	9,637	1,440	0	0	0	0	0	-9,637
Rate	7,731	1,186	0	0	4,727	667	4,727	-3,004

Source: Department of Statistics

Table 8 Evolution of exports and imports of aromatic oils of other kinds of mint (330125000)

Year	Imports: Jordanian Dinars	Imports: Quantity	National exports: JD	National exports: Quantity	Re-exported: Jordanian Dinars	Re-exported: Quantity	Total exports: Jordanian Dinar	Trade Balance (Total Exports - Imports): JD
2011	33,857	1,695	0	0	0	0	0	-33,857
2012	68,738	2,513	0	0	0	0	0	-68,738
2013	57,879	2,610	0	0	4,091	513	4,091	-53,788
2014	130,605	5,565	0	0	0	0	0	-130,605
2015	122,594	6,600	0	0	0	0	0	-122,594
Rate	82,735	3,797	0	0	818	103	818	-81,916

Source: Department of Statistics

Table 9 Development of exports and imports of other non-citrus aromaticoils (330129000)

Year	Imports: Jordanian Dinars	Imports: Quantity	National exports: JD	National exports: Quantity	Re-exported: Jordanian Dinars	Re-exported: Quantity	Total exports: Jordanian Dinar	Trade Balance (Total Exports - Imports): JD
2011	456,566	72,532	27,921	8,794	0	0	27,921	-428,645
2012	267,851	36,314	0	0	8,160	1,360	8,160	-259,691
2013	349,501	45,827	13,298	16,470	15,095	2,262	28,393	-321,108
2014	468,938	67,884	21,814	2,394	59,434	7,670	81,248	-387,690
2015	417,197	59,624	0	0	171,856	47,414	171,856	-245,341
Rate	392,011	56,436	12,607	5,532	50,909	11,741	63,516	-328,495

Source: Department of Statistics

Table 10 Development of exports and imports of other aromaticoils (330190900)

Year	Imports: Jordanian Dinars	Imports: Quantity	National exports: JD	National exports: Quantity	Re-exported: Jordanian Dinars	Re-exported: Quantity	Total exports: Jordanian Dinar	Trade Balance (Total Exports - Imports): JD
2011	149,535	74,269	9,393	32,432	8,434	627	17,827	-131,708
2012	130,828	73,180	0	0	0	0	0	-130,828
2013	215,452	133,106	6,706	1,980	0	0	6,706	-208,746
2014	191,668	102,463	18,243	36,140	12,712	933	30,955	-160,713
2015	34,417	30,615	3,000	2,250	3,379	326	6,379	-28,038
Rate	144,380	82,727	7,468	14,560	4,905	377	12,373	-132,007

Source: Department of Statistics

Table 11 Jordan's imports of aromaticoils by source for 2015

H.S CODE	Category - Country	Quantity(KG)	Value (JD)	Average Price (JD / kg)
330112000	Aromatic oils, from oranges	**16,072**	**97,880**	**6.1**
	South Africa	13,680	68,572	5.0
	India	250	7,410	29.6
	United kingdom	510	1,536	3.0
	Germany	40	1,559	39.0
	Brazil	1,592	18,803	11.8
330113000	Aromatic oils, from lemon	**1,440**	**9,637**	**6.7**
	United kingdom	1,440	9,637	6.7
330119000	Aromatic oils of other citrus	**50,603**	**137,744**	**2.7**
	Pakistan	176	2,003	11.4
	United kingdom	5,685	28,176	5.0
	Spain	34,429	79,526	2.3

	France	10,185	26,505	2.6
	Germany	128	1,534	12.0
330124000	Aromatic oils of mint	**1,000**	**18,547**	**18.5**
	India	1,000	18,547	18.5
330125000	Aromatic oils of other kinds of mint	**6,600**	**122,594**	**18.6**
	India	4,620	82,635	17.9
	PRC	1,080	21,046	19.5
	Germany	900	18,913	21.0
330129000	Other non-citrus aromatic oils	**59,624**	**417,197**	**7.0**
	Morocco	84	1,747	20.8
	India	740	20,807	28.1
	Pakistan	**43,745**	**217,925**	**5.0**
	PRC	905	11,048	12.2
	Australia	100	3,400	34.0
	United kingdom	12,960	124,787	9.6
	Hungary	200	17,383	86.9
	Italia	140	8,978	64.1
	France	500	6,937	13.9
	Germany	250	4,185	16.7
	Weighted average	**135,339**	**803,599**	**11.8**

Source: Department of Statistics

According to data from the European Union for Foreign Trade with Jordan, EU countries imported 2 tons of essential oil extracts from Jordan in 2015. Jordan imported 32 tons from the EU countries at FOB price of about 24.7 Euros per kilogram. Table (13)

Table 12 Imports and Exports of European Union Oils of aromatic Oils to Jordan

Year	Value of imports (euros)	Quantity of imports (kg)	Price of imports (EUR / kg)	Value of exports (euros)	Quantity of exports (kg)	price of exports (EUR / kg)
2011	17,374	1,000	17.4	765,068	33,000	23.2
2012	14,056	3,000	4.7	675,720	59,000	11.5

Year	Value of imports (euros)	Quantity of imports (kg)	Price of imports (EUR / kg)	Value of exports (euros)	Quantity of exports (kg)	price of exports (EUR / kg)
2013	4,061	232	17.5	479,552	54,000	8.9
2014	9,267	1,000	9.3	601,496	74,000	8.1
2015	17,925	2,000	9.0	791,836	32,000	24.7

Source :
http://exporthelp.europa.eu/thdapp/display.htm?page=st%2fst_Statistics.html&docType=main&languageId=en

Local production and consumption of medicinal and aromatic oils were estimated (Table 14) based on the study of household expenditure, which indicated that the average per capita consumption of rose water was about 90 grams per year. It was assumed that the average consumption per capita of medical and aromatic oils is about 80 grams per year and considering that the rate of consumption is linked to the population as a result of the increased demand for food, As the population growth rate in the Kingdom according to the data of the Department of Statistics is 2.2% yearly, assuming that the growth rate of demand is equal to the growth rate of population. Local demand for aromatic oils of all types was expected during (2018-2027) as shown in Table (15)

Table 13 Evolution of Imports and local production of aromatic oils (2011-2017)

The size of the Jordanian market for aromatic oils of all kinds	2011	2012	2013	2014	2015	2016	2017*
Imports (kg)	176,637	188,166	235,020	259,455	188,289	192,055	195,896
Exports (kg)	70,136	26,873	225,139	528,139	229,759	236,652	243,751
Re-exported (kg)	1,527	3,923	6,887	16,303	61,859	63,715	65,626
Local production (kg)	451,412	428,944	632,352	956,701	744,331	759,218	774,402
Total demand for goods (kg)	559,440	594,160	649,120	704,320	764,720	780,014	795,615

Source: Department of Statistics (2015), (2) Central Bank (2015)

Table 14 Estimation of the expected demand for aromatic oils (2018-2027) per ton

Demand	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Expected population (1,000 people)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Annual consumption per capita (kg / person)	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
Expected demand for aromatic oils of all types (tons)	799	816	834	853	871	891	910	930	951	972
Projected Production / Aromatic Oils (ton)	5	10	10	10	10	10	10	10	10	10
Project market share of local demand (%)	0.6%	1.2%	1.2%	1.2%	1.1%	1.1%	1.1%	1.1%	1.1%	1.0%

Source: Preparation of the team

The analysis of competitors

The medicinal and aromatic oils of medicinal plants have a great economic value as the demand for them grows locally and globally, especially with the increasing trend of the modern world to consume everything that is natural, which led to an increase in the demand of these plants, especially in the industrial and advanced countries such as the United Kingdom, This has led to higher prices, and has an economic importance and a profitable export return. (There are many fields where medicinal and aromatic plants can be used,

- Preparation of some drugs for joints pains, rheumatic infections, high blood pressure, atherosclerosis and disinfectant.
- Production of oils, where the seeds of some of these plants contain oils that are involved in the composition of some medical preparations.
- Food processing for the treatment of atherosclerosis and angina, such as the oil of the Hohoba seed, sunflower, flax, and castor.
- Preparation of cosmetics and creams of hair and soap.
- Used in the manufacturing of aromatherapy and perfumes such as roses and jasmine.
- The manufacturing of pesticides and it depends on what is found in the medicinal and aromatic plants of deadly poisons, whether for insects or fungus, such as biodegradable, Alders, henna and smoke.
- Used as spices, beverage, flavoring or aroma.

This multiplicity in uses has contributed to the creation of a large market for medicinal and aromatic plants, that its trade volume exceeds 60 billion dollars annually. Egypt, for example, is a major exporter of about 45 million dollars. The most important exporters of medicinal and aromatic plants worldwide: China - India - France - USA - Singapore - Chile. The most important exporters of medical and aromatic plants in the Middle East: Egypt - Iran - Syria - Morocco - Tunisia.

There are no statistics of the quantities of plants available in the Kingdom, which is the subject of this research, but it is found in the territory of northern Jordan and the Jordan Vallies, but in terms of extraction of essential oils locally, there is no local industry licensed, and most of what is extracted by using soaking and absorption methods by the spice dealers and specialists of popular medicine (Vegetarian), and is traded between the producer and citizen as the citizen believes that this plant or oil extracted from it are useful in the treatment of a particular disease, Without the organization of this process of production because of the absence of a specific authority to spread awareness of the benefits of these plants and their advantages, knowing that the benefits of these aromatic plants have been scientifically proven and preferred instead of chemical drugs that have side effects on the human body. The project assumes that most of its production is targeted the local markets. Currently, Jordan has three factories for the production of flavors which are used in the food industry. There are no facilities for the extraction of medicinal and aromatic oils.

The project is not expected to face difficulties in marketing its products either in the local market or in foreign markets if the production is done according to the production specifications and standards of the market.

Market share

Medical and aromatic oils from products that have a huge demand locally and internationally, therefore, marketing them through the factory or specialized exhibitions will be easy and available to the following:

-Export for other countries.

- Wholesalers.

- Local citizens (Supermarkets, grocery stores and perfumer).

Providing factories of all kinds:

- Food industry plants (as flavors and smells).
- Pharmaceutical factories.
- Perfume factories, cosmetics and beauty salons.

-Chemical factories (pesticides, soap, air fresheners, detergents, shampoos).

It is preferable to fill the resulting oils in glass containers of various shapes and sizes or containers of raw materials that do not react with the volatile oils, as well as, containers should be dark in color and have a tight lid that helps in holding volatile oils.

It is worth mentioning that packing for export in large quantities is very different from the local use in small bottles and therefore the large packagings must have good specifications and made of raw materials that do not react with volatile oils such as stainless steel and some plastics kinds used specially for chemical packaging which are tightly locked so that moisture can not be leaked to it.

Giving the current demand and the expected demand for the project's products for the coming years, it is proposed that the production capacity is about (10) tons per year, which is (10,000) liters yearly, accounting to (1.2%) of the estimated local consumption, for the above varieties referred to,, and as the required by local and external market, therefore, it's proposed that in the first year about (50%) of the proposed production capacity will be produced.

Price analysis and pricing policy

The price of a kilogram of aromatic oils is about hundreds of dinars and the prices are determined after the confirmation of quality. The price is determined according to many factors, including the confidence in the product, Ensure that restrictions and control of these products, and the availability of adequate analytical tests, natural, chemical or sensory tests.

The sale price per kilogram of the final product was estimated according to market study at 16 JD based on the prices of exports and imports of aromatic oils. However, when estimating the selling price for financial analysis, we considered that the price should be very competitive as the product will compete with other products.

A study of the market survey showed that a bottle containing (30) mm of different types of oils is priced at a minimum of 1 JD and 3 dinars for oil imported from Pakistan, where the proportion of oil is light, while the price of the package of (30 mm) of the imported oils from Turkey, ranged between (6-8) dinars based on the quality of the Turkish product and the density of oil.

Most of the local oils are not produced by local factories. Some of the traditional herbalists and specialists are involved in the production of these oils in the form of soaking and absorption. The production of aromatic oils is restricted to limited varieties of citrus and other local plants for the use in food or drink.

As for import prices from EU countries, the lowest price was 8.1 dinars / kg and the highest was 24.8 dinars / kg. As for the imported items from Pakistan, the price of one kilogram was about 5 dinars. The price of oil imported from Morocco (Argan oil) is about (20) dinars.

Note that the products produced locally are limited in the manner of soaking and absorptiononly. The project will depend on extracting aromatic oil by distillation method. The average price of a kilogram from the imported quantity is 11.87 Dinars at the C.I.F (cost, Insurance, Freight) price of imports without adding the cost of transport, clearance, customs, packaging and distribution.

Therefore, the price is 16 dinars / kg for local sale was adopted on the basis that the final price will be higher and thus increase the chance of success of the project. The proposed price is for different varieties, as the proportion of oil extraction from those varieties is low and thus, their prices are twice the estimated rate, and there are low cost categories of raw materials and thus their prices are lower than the estimated average.

Projected revenues

It is expected to produce an average of 10 tons of different types of medical and aromatic oils, 80% of which will be sold in local markets and 20% for export markets, and for the purposes of this study, the estimated average of sell price is about (16,000) dinars / ton of various aromatic oils and it is expected that the selling prices of aromatic oils will increase by 2% yearly. (Table 16)

Table 15 project revenues during the years 2018 to 2027

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total sales (kg)	**5,000**	**10,000**	**10,000**	**10,000**	**10,000**	**10,000**	**10,000**	**10,000**	**10,000**	**10,000**
Prices (JD / kg)	16.00	16.32	16.65	16.98	17.32	17.67	18.02	18.38	18.75	19.12
Total sales (JD)	80,000	163,200	166,464	169,793	173,189	176,653	180,186	183,790	187,466	191,215
Total Project revenues (JD)	**80,000**	**163,200**	**166,464**	**169,793**	**173,189**	**176,653**	**180,186**	**183,790**	**187,466**	**191,215**

3. Technical analysis

Jordan is considered rich with medicinal and aromatic plants and wild plants due to its varied climate that is suitable for the growth of these rare and important plants and it was registered more than 350 medicinal or aromatic plants. There are many medicinal and aromatic plants farmed in private farms and spread in many governorates such as Jerash and Ajloun, As well as there are in Ajloun and the northern vallies, many medicinal and aromatic plants which grow wildly. Medicinal and aromatic plants contain volatile oils and glycosides with beneficial effect, which play an important role in increasing the effectiveness of these plants.

The medicinal and aromatic plants grown in Jordan are regularly basil, mint, peppermint, aromatic oils, lemon grass, garlic, hot spices, thyme and others. It is a medicinal and aromatic plant that grows either as decorative or wildly plants such as lantana, dapella, mullet, camphor and dadora. All the extracts of plants can be used in the manufacturing of pesticides to resists plant diseases. These plants are used in the form of oils extracts, water extracts, alcohol or powder.

The extraction process of aromatic oils goes through several phases, the most important of which is sorting and grading, then the extraction phase and the phase of separation and distillation then filling. These phases include:

- Extraction phase: where the herbs are placed in the extraction machine then the temperature is being raised while passing a stream of water vapor, which leads to the extraction of aromatic oils.
- Passing a stream of steam on the condensing device, which turns the steam and oil into the liquid state and separate the oil.
- Condensing the oil again to get the greatest amount of aromatic oils.
- The extracted oil is sent to packing.

Experiments were conducted at the National Center for Agricultural Research and Extension (NARCE) on the appropriate temperature, appropriate distillation time and quantity of distilled oils for various medicinal and aromatic plants in Jordan. Table 17 shows the percentage of oil extracted per millimeter per kilogram of medicinal and aromatic plants. This table will be used to estimate the required quantities of raw materials.

Table 16 The amount of oil distilled for different types of medicinal and aromatic plants

Plant	Time (min)	Temperature	Ml/kg
Melissa	50	140	1.5-2
Lavender	60	150	2-3

Plant	Time (min)	Temperature	Ml/kg
Lemon Grass	30	130	4.5-5
Hossalban	45	140	2
Kina	60	160	3-4
Giranium	40	150	2.5-3
basil	25	120	0.5-1
Schumer	30	120	1
Thyme	90	180	2-4
Mérimée	90	170	3.5-4

Source: National Center for Agricultural Research and Extension

These oils are not supplied and extracted 100%, as some of them are volatile oils, so it is necessary to add a carrier or inert or diluted with alcohol or propyl glycol or oils with good concentrations that make products to be handled and preferable by the consumer and beneficial to the owner of the project. As 100% of the pure extractions will be expensive, and sometimes will be difficult trading with non-specialists. The ratio of active ingredient and concentration in the packaging varies from product to another and can be addressed in the detailed feasibility study.

The required raw materials for the project

Many supplies are required for the production of varios kinds of aromatic oils, in addition to the raw materials from which oil is extracted, there is a need for inert materials from propylene glycol oils, fillers, cartons and stickers.

The following table shows the cost of raw materials to produce the required volume of aromatic oils.

Table 17 Raw materials

Substance	Quantity (kg)	Price JD/kg	Cost JD	Substance	Quantity (kg)	Price JD/kg	Cost JD
Laurel paper	1000	0.5	500	Apricot seeds	1000	3	3000
Chamomile	500	3	1500	Anise seeds	1000	5	5000
Onions	500	0.25	125	Propyl Glycol	7000	2.5	17500
parsley	2000	0.2	400	Cardamom	1000	5	5000
Al Banafsaj	500	4	2000	Black seeds	2000	2	4000

Watercress	500	0.2	100	Lemon Grass	2000	0.3	600
Aldjadh	500	0.15	75	Hossalban	1000	1	1000
Fenugreek	500	0.25	125	Poppy flower	500	2	1000
Al Khosama	1000	1	1000	Olive oil	1000	5	5000
Basil	500	0.5	250	Inert oils	4000	2	8000
Shumer	500	0.25	125	Garlic cloves	1000	1	1000
Wormwood	500	0.2	100	Citrus peel	1000	0.18	180
Nettles	500	0.15	75	Poppy husks	1000	0.5	500
Mérimée	1000	0.25	250	Mint leaf	1000	0.25	250
Mint	500	0.2	100	Poppy leaf	1000	0.5	500
Aniseed	500	2	1000	100ml dark glass jars	111,800	0.04	4472
Al Khorfaish seeds	500	2	1000	Carton trays	10,525	0.07	736.75
Sweet almond seeds	1000	3	3000	Label and Posters	111,800	0.007	782.6
Bitter almond seeds	1000	2	2000	Total			71463.75

The required human resources

The production of aromatic oils of all kinds requires four unskilled workers working together for one shift per day, in addition to an agricultural engineer, specialized in plant production, or food processing or chemical, who is fully aware of the production of aromatic oils. And one person is enough to run this project, and assume that the project manager may devote his time to the deal with companies and factories wanting to purchase the product as well as marketing the production and to negotiate the purchase of medicinal and aromatic plants, and may need a second person (supervisor production) for field supervision on the productionand distillation process and marketing aromatic oils with commission, the following table shows required labor:

Table 18 the required labor for the project

Job title	No.

Job title	No.
Indirect staff	
Project Manager / Plant Production Engineer	**1**
Production Supervisor	**1**
Marketing Agents	**1**
Total Non-Direct staff	**3**
Manufacturing staff	2
Handling and loading	2
Total Direct staff	4
Grand Total	7

- **Agricultural Engineer:** a good knowledge of how to manufacture essential oils of all kinds and how to treat them. Ability to request and provide the necessary raw materials and tools and supplies of production process, such as seeds, leaves, oils required and supervision of the manufacturing process during the preparation phase. Distillation, preparation and packaging, as well as adjusting the conditions of distillation process according to instructions. Draw samples to be analyses in the laboratory, observation of the distillation process, and diagnosis of deviations in measurements and distillates results. Inspect the conformity of the product to the required specifications, filling and working models, documenting expenditures, proposing policies and work plans, following up on their implementation and evaluating their results. Organizing daily work, distributing workers among tasks, following up on their work performance. Ensure the safety of the supply of raw materials and production.
- **Production supervisor:** inspect the equipment and equipment of the essential oils manufacturing unit, such as: extraction and distillation apparatus, heating and cooling systems and temperature measuring devices. Check the plants, flowers and / or seeds to be extracted, processed, placed in the receptacle of the extractor, and add solvent or water vapor to the extraction device. Operate and monitor the extraction device according to the manufacturer's instructions, and operate the distillation device and its accessories. Feed it with the extracted oil mixture and solvent, monitor it according to the manufacturer's instructions, and collect the oils and separated from the solvent. Draw samples to be

analyzed in the laboratory, and the implementation of necessary corrective actions. Service and equipment of the essential oil manufacturing unit, cleaning machine and cleaning the workplace. Registration of quantities and types of consumables and quantities of oils produced and concentrations. Application procedures and instructions of the occupational safety and health.

- **Manufacturing Staff**: Receiving raw materials such as citrus peel, wild flowers, thyme, lemon grace, salve etc. weight, and follow the specifications. Put the raw material inside the boiler, adjust the amount of water, the amount of steam and the temperature needed for the distillation process, and run the boiler. Introducing the raw material in specified quantities in boilers, monitor distillation process, diagnosis of deviations in measurements and distillation products, and control of factors affecting quality, such as temperature, pressure and quantity of cooling water. Adjust the distillation process as needed. Dump waste distillation and cleaning boilers. Mobilization of business models. Application procedures and instructions of the occupational safety and health.

Site analysis

It is proposed that the project site is located in Ajloun governorate and close to its agricultural areas, where the project needs to collect plants and contract with farmers to cultivate plants in rainfed and irrigated lands for the production of the required aromatic and medicinal plants.

Accordingly, it is proposed that the factory for the distillation of medical and aromatic oils will be established in Ajloun Governorate, so that the suitable location for production will be chosen based on the availability of infrastructure services such as water and electricity and the availability of raw materials.

Description of the production process and manufacturing steps

Due to the very huge variation in the composition of different aromatic oils, it must be obtained from different plants and in ways that suit the nature of each plant and the quality of aromatic oils and its proportion and degree of sensitivity to heat and its uses. Therefore, the composition of the aromatic oils produced vary depending on the nature of each plant and its natural specifications, The product of aromatic oils is important and very sensitive that must be well taken care of, free of exotic sediments and free of toxic metals and non-volatile substances to use these oils in important and precise industries such as pharmaceuticals, perfumes, cosmetics and food industries. Due to the importance and seriousness of the use of these products, Jordan Standards and Metrology Organization had issued the standard specifications for aromatic oils - lemon oil produced by pressing No. (960) for the year 2007 and the standards of aromatic water No. (740) for the year 2014, which emphasizes the importance of tests on the final product, including:

1. Chemical tests (Ester number and pH number).
2. Natural tests (specific weight estimation, virtual density estimation, refractive index estimation, alcohol solubility estimate).

Manufacturing phaseses

The process of manufacturing aromatic and medical oils and extract them from plants passes through several stages, including:

The First Phase: The cleaning phase

Medicinal and aromatic plants and seeds are cleaned thoroughly of dust, insects and exotic plants that may exist with native plants and all other types of impurities.

The plants must be aromatic and with natural smell and color, and free from clotting and blocs. The process of cleaning, sorting, purification, washing with running water and then filter the water.

Second Phase: Processing Phase

The plants are either sliced into pieces or sliced into large slices, and the branches can be sliced into thin strips or shredded for small pieces. Such as (lemon, orange).

Third Phase: Freezing Phase

Plants, especially the parts of flowers, leaves, grasses and small branches, are placed directly in the oil extraction apparatus. The other parts, which are grinded, grated and crushed (the large parts) are placed before the extraction process in a refrigerator under 20 ° C for two or three days, then enter the extraction process while thry are still freezed.. The process preserves ingredients as they are and in good condition such as (roses, mint, basil)

Fourth Phase: Distillation process

The process of distillation is the essential step in the extraction of medicinal and aromatic oils. The distillation process is divided into three basic types depending on the plant type, sensitivity and heat tolerance.

1. Water distillation

This method is used for medicinal and aromatic plants that bear a temperature slightly above 100° C, which is the boiling point of water. Such as (almonds, hazelnuts)

2. Indirect steam distillation

This method is suitable for plants containing aromatic oils that can not withstand a temperature exceeds 100 ° C, in which the vapor generated outside the device is passed directly into the water. Such as rose, basil, mint.

3. Direct steam distillation

This method is suitable for fresh, non-dried plants and uses steam by passing it directly on the plants to extract its oils.

Figure (3) illustrates the principle of distillation device and shows the possibility of using the same device for distillation in the previous three methods depending on the type of plants required to extract their oils by changing the method of work either by using water directly or by passing the steam into the water or using steam directly on the plants.

Method of operating the distillation device:

(A) Device components

1. Wire basket (stainless steel) in the shape of a cylinder.
2. Double walled cylindrical bowl made of stainless steel. Provided with a bottom openining to filter the water after the distillation process. And a high side opening on which the condenser that receives the oil is installed.
3. Sources of heating either by the hot water inside the container, or the source of direct steam entry from an external source within the water or only source of steamover the plants in the basket inside the container.
4. Source to enter the cold water that helps the condenser to perform its task in intensifying oils and an exit hole for cooling water.
5. A vessel to receive oil and separates it from water
6. Absorbent pump: to re- distill the resulting distillation water to extract all existing oils.
7. The cover is made of the same raw material as the container. It is equipped with a thermometer for measuring temperature, a bottle and a pressure gauge.

(B) Method of operating the distillation device:

1. The basket filled with the required plants to obtain its volatile oils.

2. The lid of the distillation unit is closed.

3. In the case of distillation with water or indirect steam, the machine should be supplied with water.

4. In the case of distillation with indirect steam, there is no need for water.

5. Vaporized steam carries the volatile oil and the intensify phase occurs in the condenser.

6. Always take into account that the steam hole is directly proportional to the condensor capacity so that the vapors do not leak without condensation. The temperature of cooling water outside the condenser should not exceed 30 ° C.

7. The time needed is calculated by experience. Where each specific amount of a particular plant at a certain time is sufficient to obtain the existing oil.

8. The resulting distilled water shall be treated again to extract all the possible oil, indicating the color mixed with the water.

9. All connections should be closed and also for condenser.

10. The machine capacity is 50 liters, 25 kg.

The manufacturing process goes through the following stages:

- The plants are separated by a machine with knives that move up and down while the basin moves in a circular way.
- The chopped plants are sprayed with a harvester machine (like a grain milling machine) containing a spiral rotating slowly in the body of the mash machine.
- The powder is mixed by 10-30% Carrier Oil (sunflower oil) to activate the process of extracting oil, and then transferred to the distillation steam, to fill the oil extracted from there to external bottles and then to cardboard boxes and then to the warehouse.

Note that plants vary by type, giving between 0.3% and 4.5% pure aromatic oil for direct consumption, for the required time, and to stabilize this oil and increase the proportion extracted economically, while at the same time, this oil is converted to direct consumption for the required time period. The plant powder is mixed with Carrier Oil, such as sunflower oil or corn, between 10% and 30%, and then converted to steam distillation. Thus, aromatic oil that is ready for use can be obtained by 10%.

In terms of the operation of steam distillation apparatus, the illustration shows the scientific principle of the steam distillation system, where a stream of water vapor passes through aromatic plants grinded and heated in a closed tank, then the oil evaporates and carries steam to the cold condenser.

The oil and water condense, then the oil is separated from the water in a pot after the oil floats above the surface of the water because it is less dense and does not dissolove in it, and then returned to the tank again. Note that organic solvents (depending on the plant type) can be used instead of water in distillation to extract aromatic oil. Although the statistics show the quantities of aromatic oils by weight, however, because the product is liquid, we will assume for the purposes of study to consider the specific density of oil is (1) and therefore consider the production capacity is about (10,000) liters.

The analysis of technical requirements

1.1.1 Land and cunstruction

It is possible to buy 500 m2 of land for the establishment of thefactory, and it's possible to rent a building or a metal hangar with an area of 500 m2. The cost of such a building in the Ajloun governorate is estimated at (6,000 JD).

Table19 the required land for the project

Land	Cost (JD/sq.m)	Area sq.m
The required land	**25**	500

The following table shows the required construction work for the project:

Table 20 the required construction work

Construction work	Cost (JD/sq.m)	**Area (m^2)**
Administrative offices	50	50
Raw materials warehouse	100	100
cooled Warehouse	10	10
Room for workers	50	50
Packing workshop	100	100
Bathrooms and restroom	70	70
Total	**380**	

Machinery and Equipment

The following table shows the required machinery and equipment needed for the project:

Table 21 Machinery and equipments

Machinery and equipment	Quantity	Cost (JD per unit)
Punching machine for raw materials and plants	1	1,000
Milling machine for plants	1	1,500
Hot water and steam pipes to distillation device	1	1,000
Steam distillation apparatus	1	5,000
Filling machine with suction pump	1	2,000
Spiral mill for herbs	1	2,000
Chakushian Mill for Grass, Stainless	1	2,500
20 liter / hour steam distillation device, stainless steel	1	2,000
Water cooling unit for condenser with 2 hp	1	1,000
Stainless Tables	3	250
30 liter stainless steel extractors	20	50
Refinery distillation unit 10 liter / hour	1	1,500
Stainless steel rotary extraction device of 20 liter	1	2,000
Balance 100 kg	1	150
Laboratory tools	1	500
Sensitive Balance	1	250
Freezer	1	1,000
Refrigerator	1	600
Flasks circular carrier mechanically managed in a circular	1	2,000
Packaging unit	1	500
Different tools	1	500

Furniture

The project needs the following furniture:

Table 22 the required furniture for the project

	Quantity	Unit cost / JD
Office Furniture	1	1,000
Computer and software	1	1,000
Kitchen tools and cafeteria	1	1,000

Vehicles and Transportation

The following table shows a list of vehicles and means of transportion required for the project

Table 23 Vehicles and Transportation

Vehicles and Transportation	Quantity	Unit cost (JD)
Pickup truck	1	15,000

4. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.97% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Icome tax on the net project income is 0 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 24 General Assumptions

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 25 Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.705
Euro	0.754

Table 26 Annual Growth Rates Assumptions

Annual Growth Rates	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 27 Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	0.00%
Utilities Expenses from Total Revenues	1.00%
Maintenance Expenses from Total Revenues	0.05%
Depreciation Expenses from Total Revenues	0.05%
Insurance Expenses from Total Assets Value	0.00%
Marketing Expenses from Total Revenues	2.00%

Table 28 Income Tax Assumptions

Income Tax Assumptions	Value
Average Income Tax in Jordan(1)	5%
Income Tax Deduction(2)	100%
Income Tax after Deduction	0%
Compulsary Reserve Percentage(3)	10%
Other Assumptions	**Value**
Annual Monthly Salaries have been calculated after multiplying monthly salaries	16

Risk Premuim	
Risk Free Rate of Return (1)	6.50%
Return to Debt Maturity (2)	7.50%
Market Risk Premuim (3)	9.93%
Income Tax Rate	0.00%
Beta (4)	1
Growth rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 29 : Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premuim	%9.93
Income Tax Rate	%0
Bets	1.00
Equity	63,171
Loans Value	63,171
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.50
Loans Value	63,171
Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	63,171
Equety Percentage	%50
Gross	
Project Value	126,342
Proportion of the project	%100
Weighted Average Cost of Capital	**%11.97**

Capital Expenditures

The estimated cost of the project is 114,857 JD and it covers land costs, construction works, machinery and equipment, furniture, pre-operating expenses and working capital. The following tables summarize the details of these expenses:

Table 30 Land Capital Expenditure

Land	(Jordan Dinar/Meters Square) Cost	Area (M2)	Cost
Required Land	25	500	12,500
Cost			**12,500**

Table 31 Construction Work Capital Expenditure

Description	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost Jordan /
Administrative offices	50	50	2,500
Raw materials warehouse	100	100	10,000
cooled Warehouse	10	10	100
Room for workers	50	50	2,500
Packing workshop	100	100	10,000
Bathrooms and restroom	70	70	4,900
Other works	-	-	2,510
Total	**380**		**32,510**

Table 32 Machinery and Equipment Capital Expenditure

Machinery and equipment	Quantity	Cost	Total
Punching machine for raw materials and plants	1	1,000	1,000
Milling machine for plants	1	1,500	1,500
Hot water and steam pipes to distillation device	1	1,000	1,000
Steam distillation apparatus	1	5,000	5,000
Filling machine with suction pump	1	2,000	2,000
Spiral mill for herbs	1	2,000	2,000
Chakushian Mill for Grass, Stainless	1	2,500	2,500
20 liter / hour steam distillation device, stainless steel	1	2,000	2,000
Water cooling unit for condenser with 2 hp	1	1,000	1,000
Stainless Tables	3	250	750
30 liter stainless steel extractors	20	50	1,000
Refinery distillation unit 10 liter / hour	1	1,500	1,500

Machinery and equipment	Quantity	Cost	Total
Stainless steel rotary extraction device of 20 liter	1	2,000	2,000
Balance 100 kg	1	150	150
Laboratory tools	1	500	500
Sensitive Balance	1	250	250
Freezer	1	1,000	1,000
Refrigerator	1	600	600
Flasks circular carrier mechanically managed in a circular manner	1	2,000	2,000
Packaging unit	1	500	500
Different tools	1	500	500
Total (JD)			**28,750**

Table 33 Furniture Capital Expenditure

Furniture and furnishings	Quantity	Cost / JD	Total Cost
Office Furniture	1	1,000	1,000
Computer and software	1	1,000	1,000
Kitchen tools and cafeteria	1	1,000	1,000
Total			3,000

Table 34 Transporation and Vehicles Capex

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Pick up	1	15,000	15,000
Total			**15,000**

Table 35 Pre Operating Expenses

Pre-operating expenses	Estimaited cost (JD)
Governmental fees	1,000
Transportation	500
Marketing	2,000
Training	1,000
Total	**4500**

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 36 Working Capital

Working Capital	No. of Months	%	Estimated Cost (JD)
Operating expenses	3	%25	13,489
General and administrative expenses	3	%25	238
Marketing expenses	3	%25	270
Indirect Salaries	3	%25	4,600
Total			**18,597**

Table 37 Expenses Summary

Capital expenditure summary	Cost (JD)	Contingency	Cost (JD)	%
Land	12,500	10%	13,750	%10.9
Construction works	32,510	10%	35,761	%28.3
Machinery &Equipment	28,750	10%	31,625	%25.0
Furniture	3,000	10%	3,300	%2.6
Vehicles	15,000	10%	16,500	%13.1
Pre-Operating Expenses	4,500	10%	4,950	%3.9
Working capital	18,597	10%	20,456	%16.2
Total (JD)	**114,857**		**126,342**	**100%**

Sources of Funding

The financing structure of the project consists of loans and Equity. The investment cost of this project is estimated at (126,342) JD, where 50% of the project will be financed through loans (63,171) JD, and the remaining (50%) through (Equity) with (63,171) JD. The following table summarizes the general structure of the required financing:

Table 38 Sources of Funding

Funding Sources	Amount	Percentage
Equity (Self Financing)	63,171	%50
Loans	63,171	%50
Total	**126,342**	**%100**
Use of Fund	**Value**	**Ratio**
Capital Expenditures	100,936	%79.89

Pre-Operating Expenses	4,950	%3.92
Working Capital	20,456	%16.19
Total	**126,342**	**%100.00**

Operating Expenses

The following table sumarizes the expected operating expenses for the proposed project (2018-2027):

Table 39 Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw Materials and water	35,732	71,464	72,893	74,351	75,838	77,355	78,902	80,480	82,089	83,731
Utilities	40	82	83	85	87	88	90	92	94	96
Maintenance	40	82	83	85	87	88	90	92	94	96
Disposables	40	82	83	85	87	88	90	92	94	96
Direct Salaries	13,200	18,128	18,491	18,860	19,238	19,622	20,015	20,415	20,823	21,240
Others	4,905	8,984	9,163	9,347	9,534	9,724	9,919	10,117	10,319	10,526
Total	**53,957**	**98,820**	**100,797**	**102,813**	**104,869**	**106,966**	**109,105**	**111,288**	**113,513**	**115,784**

General and Administrative Expenses

The following table sumarizes the expected general and administrative expenses for the proposed project (2018-2027):

Table 40 General and Administrative Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Telecom	200	204	208	212	216	221	225	230	234	239
Hospitality	100	102	104	106	108	110	113	115	117	120
Staionary and Printing	50	51	52	53	54	55	56	57	59	60

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Transportaton	100	102	104	106	108	110	113	115	117	120
Miscellaneous	500	46	47	48	49	50	51	52	53	54
Total	**950**	**505**	**515**	**525**	**536**	**547**	**557**	**569**	**580**	**592**

Marketing Expenses

Table 41 Marketing Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	**1,079**	**1,976**	**2,016**	**2,056**	**2,097**	**2,139**	**2,182**	**2,226**	**2,270**	**2,316**

Human Resources

The total annual salaries and monthly wages has been estimated on 16 months for the purpose of considering annual increments on salaries as well as payments related to social security and health insurance,... etc.

The following table summerizes the expected annual salaries and wages, considering annual increment on salaries as per the income statement:

Table 42 Expected monthly salaries (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
Project Manager / Plant Production Engineer	600	618	637	656	675	696	716	738	760	783
Production Supervisor	450	464	477	492	506	522	537	553	570	587
Marketing Agents	100	103	106	109	113	116	119	123	127	130

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Direct staff										
Manufacturing staff	300	309	318	328	338	348	358	369	380	391
Handling and loading	250	258	265	273	281	290	299	307	317	326

Table 43 Annual Expected Salaries

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect staff										
Project Manager / Plant Production Engineer	9,600	9,888	10,185	10,490	10,805	11,129	11,463	11,807	12,161	12,526
Production Supervisor	7,200	7,416	7,638	7,868	8,104	8,347	8,597	8,855	9,121	9,394
Marketing Agents	1,600	1,648	1,697	1,748	1,801	1,855	1,910	1,968	2,027	2,088
Total Indirect	**18,400**	**18,952**	**19,521**	**20,106**	**20,709**	**21,331**	**21,971**	**22,630**	**23,309**	**24,008**
Direct staff										
Manufacturing staff	7,200	9,888	10,185	10,490	10,805	11,129	11,463	11,807	12,161	12,526
Handling and loading	6,000	8,240	8,487	8,742	9,004	9,274	9,552	9,839	10,134	10,438
Total Direct	**13,200**	**18,128**	**18,672**	**19,232**	**19,809**	**20,403**	**21,015**	**21,646**	**22,295**	**22,964**
Grand Total	**31,600**	**37,080**	**38,192**	**39,338**	**40,518**	**41,734**	**42,986**	**44,275**	**45,604**	**46,972**

Depreciations

The following tables illustrate cost of Capex and Depreciaition rates:

Table 44 Capex and Depreciation Expenses

Capex Cost and Annual Depreciation Rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	13,750	%0.0	%0.0
Construction Works	35,761	%5.0	%0.0
Machinaries	31,625	%10.0	%2.0
Furniture	3,300	%10.0	%5.0
Vehicles	16,500	%10.0	%0.0
	100,936		

Table 45 Depreciations and Additions on Construction Works

Capex, Annual Additions and Depreciaitions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Fixed Assets	100,936	101,734	102,552	103,392	104,254	105,139	106,048	106,982	107,940	108,925
Total Depreciation Expenses	6,931	7,010	7,092	7,176	7,262	7,351	7,442	7,535	7,631	7,729
Total Accumulated Depreciation	6,931	13,941	21,033	28,209	35,471	42,822	50,264	57,799	65,430	73,160
Total Additions	0	798	818	840	862	885	909	933	959	985
Total Net Book Values	94,005	87,793	81,519	75,183	68,783	62,317	55,784	49,182	42,510	35,765

Loan

The table below summarizes all details related to the loan such as annual installments and payment methods for the remaining amount of the loan for each year of the project.

Table 46 Loan Details

Loan Value	193,362
Annual Interst Rate	9.50%
Loans Period	5
The grace period	1
Loan Starts at year:	2018

Year	Annual Payment	Interest	Capital	Loan Remaining Value
2018				63,171
2019	16,452	6,001	10,451	52,720
2020	16,452	5,008	11,444	41,277
2021	16,452	3,921	12,531	28,746
2022	16,452	2,731	13,721	15,025
2023	16,452	1,427	15,025	0

Income Statement

Table 47 Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	80,000	163,200	166,464	169,793	173,189	176,653	180,186	183,790	187,466	191,215
Gross Operating Revenues	80,000	163,200	166,464	169,793	173,189	176,653	180,186	183,790	187,466	191,215
Operating Expenses	(53,957)	(98,820)	(100,797)	(102,813)	(104,869)	(106,966)	(109,105)	(111,288)	(113,513)	(115,784)
Gross Operating Profit	26,043	64,380	65,667	66,981	68,320	69,687	71,080	72,502	73,952	75,431
Gross Profit Percentage	*%33*	*%39*	*%39*	*%39*	*%39*	*%39*	*%39*	*%39*	*%39*	*%39*
Salaries and Benefits (Indirect Staff)	(18,400)	(18,952)	(19,521)	(20,106)	(20,709)	(21,331)	(21,971)	(22,630)	(23,309)	(24,008)
General and Administraive Expenses	(950)	(505)	(515)	(525)	(536)	(547)	(557)	(569)	(580)	(592)
Markeing Expenses	(1,079)	(1,976)	(2,016)	(2,056)	(2,097)	(2,139)	(2,182)	(2,226)	(2,270)	(2,316)
Pre Operating Expenses	(4,950)									
Gross Indirect Expenses	(25,379)	(21,433)	(22,051)	(22,688)	(23,343)	(24,016)	(24,710)	(25,424)	(26,159)	(26,915)
Income before Interest, Depreciation and Tax	664	42,946	43,616	44,293	44,978	45,670	46,370	47,078	47,793	48,516
Fixed Assets Depreciations	(6,931)	(7,010)	(7,092)	(7,176)	(7,262)	(7,351)	(7,442)	(7,535)	(7,631)	(7,729)
Income before Tax and Interests	(6,267)	35,936	36,524	37,117	37,715	38,319	38,929	39,543	40,162	40,787
Bank Interests	(6,001)	(5,008)	(3,921)	(2,731)	(1,427)	0	0	0	0	0
Income Before Tax	(12,268)	30,928	32,602	34,386	36,288	38,319	38,929	39,543	40,162	40,787

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	(12,268)	30,928	32,602	34,386	36,288	38,319	38,929	39,543	40,162	40,787
Net Profit Percentage	*%15-*	*%19*	*%20*	*%20*	*%21*	*%22*	*%22*	*%22*	*%21*	*%21*
Compulsory Reserves	0	(3,093)	(3,260)	(3,439)	(3,629)	(3,832)	(3,893)	(3,954)	(4,016)	(4,079)
Retained Earnings	(12,268)	15,567	44,909	75,857	108,516	143,003	178,039	213,628	249,774	286,482

Expected Cashflows Statement

The table belwo summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 48 Expected Cashflows

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash flows from operating activities										
Net profit (loss)	(12,268)	30,928	32,602	34,386	36,288	38,319	38,929	39,543	40,162	40,787
Non-monetary adjustments										
Bank benefits	6,001	5,008	3,921	2,731	1,427	0	0	0	0	0
Consumption	6,931	7,010	7,092	7,176	7,262	7,351	7,442	7,535	7,631	7,729
Operating cash flows before changes in working capital	664	42,946	43,616	44,293	44,978	45,670	46,370	47,078	47,793	48,516
Shortage (increase) in inventory	(2,978)	(2,978)	(119)	(121)	(124)	(126)	(129)	(132)	(134)	(137)
Decrease (increase) in receivables	(6,667)	(6,933)	(272)	(277)	(283)	(289)	(294)	(300)	(306)	(312)

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
(Decrease) increase in accounts payable	1,519	761	46	47	47	48	49	50	51	52
Change in working capital	(8,126)	(9,150)	(346)	(352)	(359)	(367)	(374)	(381)	(389)	(397)
Net cash flows (used) from operating activities	(7,462)	33,796	43,270	43,941	44,618	45,304	45,996	46,697	47,404	48,119
Cash flows from investing activities										
(Purchase) of fixed assets	(100,936)	(798)	(818)	(840)	(862)	(885)	(909)	(933)	(959)	(985)
Net flows (used) from investment activities	(100,936)	(798)	(818)	(840)	(862)	(885)	(909)	(933)	(959)	(985)
Cash flows from financing activities										
Equity capital	63,171									
Loan Amortization	(10,451)	(11,444)	(12,531)	(13,721)	(15,025)	0	0	0	0	0
Bank Interest Rate	(6,001)	(5,008)	(3,921)	(2,731)	(1,427)	0	0	0	0	0
Loans	63,171	0	0	0	0	0	0	0	0	0
Net cash flows (used) from financing activities	109,890	(16,452)	(16,452)	(16,452)	(16,452)	0	0	0	0	0
Net increase (decrease) in cash	1,492	16,547	26,000	26,649	27,304	44,418	45,087	45,763	46,446	47,134
Cashflows at the Beginning of Period	0	1,492	18,039	44,039	70,688	97,992	142,410	187,498	233,261	279,706
Cashflows at the End of Period	1,492	18,039	44,039	70,688	97,992	142,410	187,498	233,261	279,706	326,841

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 49 Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	1,492	18,039	44,039	70,688	97,992	142,410	187,498	233,261	279,706	326,841
Inventory	2,978	5,955	6,074	6,196	6,320	6,446	6,575	6,707	6,841	6,978
Accounts receivable	6,667	13,600	13,872	14,149	14,432	14,721	15,015	15,316	15,622	15,935
Total assets traded	**11,137**	**37,595**	**63,986**	**91,033**	**118,744**	**163,578**	**209,088**	**255,283**	**302,169**	**349,753**
Non-current assets										
Fixed assets (net)	94,005	87,793	81,519	75,183	68,783	62,317	55,784	49,182	42,510	35,765
Total non-current assets	**94,005**	**87,793**	**81,519**	**75,183**	**68,783**	**62,317**	**55,784**	**49,182**	**42,510**	**35,765**
Total assets	**105,142**	**125,387**	**145,505**	**166,216**	**187,527**	**225,894**	**264,872**	**304,466**	**344,679**	**385,518**
Liabilities	-	-	-	-	-					
Current liabilities										
Payables	1,519	2,280	2,325	2,372	2,419	2,468	2,517	2,567	2,619	2,671
Remaining amount of Loan	11,444	12,531	13,721	15,025	0	0	0	0	0	0
Total current liabilities	**12,962**	**14,810**	**16,046**	**17,397**	**2,419**	**2,468**	**2,517**	**2,567**	**2,619**	**2,671**
Non-current liabilities										

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
long term loans	41,277	28,746	15,025	0	0	0	0	0	0	0
Total long - term liabilities	**41,277**	**28,746**	**15,025**	**0**	**0**	**0**	**0**	**0**	**0**	**0**
Total liabilities	**54,239**	**43,556**	**31,071**	**17,397**	**2,419**	**2,468**	**2,517**	**2,567**	**2,619**	**2,671**
Property rights										
Member Contributions	63,171	63,171	63,171	63,171	63,171	63,171	63,171	63,171	63,171	63,171
Compulsory reserve	0	3,093	6,353	9,792	13,420	17,252	21,145	25,100	29,116	33,194
Profit (loss) round	(12,268)	15,567	44,909	75,857	108,516	143,003	178,039	213,628	249,774	286,482
Total equity	50,903	81,831	114,433	148,819	185,107	223,427	262,355	301,898	342,061	382,847
Total liabilities and equity	**105,142**	**125,387**	**145,505**	**166,216**	**187,527**	**225,894**	**264,872**	**304,466**	**344,679**	**385,518**

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 50 Free Net Cash flows Table

Net Cashflows	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final value
Net Free Cashflows	(126,342)	(2,512)	32,999	42,452	43,101	43,756	44,418	45,087	45,763	46,446	47,134	615,440

Discount Factor	1.00	0.89	0.80	0.71	0.64	0.57	0.51	0.45	0.40	0.36	0.32	0.32
Net Present Value for Cash	**(126,342)**	**(2,243)**	**26,323**	**30,245**	**27,425**	**24,867**	**22,546**	**20,440**	**18,529**	**16,796**	**15,224**	**198,775**

Table below (51) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates a total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 51 payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	(126,342)	(2,512)	32,999	42,452	43,101	43,756	44,418	45,087	45,763	46,446	47,134
Project Value Returned	126,342	128,854	95,855	53,403	10,302	(33,454)	(77,872)	(122,960)	(168,723)	215,168)	(262,303)
Payback Period (Year)	4	1	1	1	1	0	0	0	0	0	0

Table 52 Financial Analysis Results

Weighted Average Cost of Capital (WACC)	**%11.97**
Net Present Value for Cashflows	**272,585**
Payback Period	**4**
Internal Rate of Return	**%30.26**

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project inrernal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 272,585 Jordan Dinars considering that the project provides 7 Job Opportunitites for the governorate residents.

Table 53 Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	Net present value (Thousand JD)	WACC
Original Scenario	**30.26%**	**4**	**273**	**11.97%**
Revenues declined by 10%	**19.75%**	**6**	**103**	**11.97%**
Operating Expenses Increased by 10%	**23.61%**	**5**	**163**	**11.97%**
Total cost increased 10%	**22.00%**	**5**	**150**	**11.97%**
Costs increased and declined by 10% togather	**9.45%**	**10**	**-19**	**11.97%**